

LAB VIVA PATTERN

Subject: Linked Lists, Stacks, Queues

Paper: Lab Viva Pattern - Set A | Duration: 3 Hours | Max Marks: 72 | Date: 11-03-2026

Total Questions: 21 | Answer all questions.

Section 1: MCQ (11 Questions - 22 Marks)

Q1. [2 marks] (Medium)

Here are the 20 unique exam questions for your Lab Viva Pattern exam:

Q2. [2 marks] (Easy)

Which of the following is a non-linear data structure?

Options:

A) Array B) Linked List C) Stack D) Tree

Q4. [2 marks] (Easy)

In a stack, which operation removes an element from the top?

Options:

A) push B) pop C) peek D) enqueue

Q6. [2 marks] (Medium)

What is the state of a stack if, after pushing 5, 10, 15 and then popping twice, you push 20?

Options:

A) [5, 15, 20] B) [5, 10, 20] C) [5, 20] D) [20, 15, 5]

Q8. [2 marks] (Medium)

If a linked list has 5 nodes and you insert a new node at the beginning, how many nodes are traversed?

Options:

A) 0 B) 1 C) 5 D) 6

Q10. [2 marks] (Easy)

Which data structure uses the enqueue and dequeue operations?

Options:

A) Linked List B) Stack C) Queue D) Tree

Q12. [2 marks] (Hard)

A recursive function calls itself. Which data structure is implicitly used by the system to manage these function calls?

Options:

A) Queue B) Heap C) Stack D) Hash Table

Q14. [2 marks] (Easy)

What does the NULL pointer typically signify in a linked list?

Options:

A) The first node B) An error C) The end of the list D) A deleted node

Q16. [2 marks] (Hard)

Which of the following operations has the highest time complexity in a singly linked list of 'N' nodes?

Options:

A) Deleting the first node B) Inserting a node at the beginning C) Accessing the Nth node D) Deleting the last node

Q18. [2 marks] (Medium)

What would be the output if you pop an empty stack?

Options:

A) A random value B) 0 C) An error/exception D) Nothing happens

Q20. [2 marks] (Medium)

In a singly linked list, if head points to the first node, how do you access the data of the second node?

Options:

A) head->next->data B) head.next.data C) head->data->next D) head[1].data

Section 2: Short Answer (10 Questions - 50 Marks)

Q3. [5 marks] (Medium)

Explain the primary advantage of using a linked list over an array for dynamic data storage.

Q5. [5 marks] (Hard)

Describe a scenario where a doubly linked list would be more efficient than a singly linked list.

Q7. [5 marks] (Easy)

What is the fundamental principle on which a queue operates?

Q9. [5 marks] (Hard)

Explain how a stack can be used to check for balanced parentheses in an expression.

Q11. [5 marks] (Medium)

Differentiate between static and dynamic memory allocation in the context of linked lists versus arrays.

Q13. [5 marks] (Medium)

Describe a situation where a circular queue would be more beneficial than a linear queue implemented using an array.

Q15. [5 marks] (Medium)

Explain how an array can be used to implement a stack. What are its limitations?

Q17. [5 marks] (Easy)

Name two common applications of queues.

Q19. [5 marks] (Hard)

How can two stacks be efficiently implemented using a single array such that neither stack overflows until the entire array is full?

Q21. [5 marks] (Hard)

Explain the difference between an ordinary queue and a priority queue.

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